

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Physics

- 1 A solid block of uniform cross-sectional area A and mass m floats upright in a liquid of density ρ_0 , as shown in the diagram. The lower portion of the block is submerged to a depth h .

The liquid is then heated uniformly. The liquid expands so that its density ρ_0 decreases, while the thermal expansion of the solid block is negligible. The block remains floating upright.

Here are three statements about the system after the liquid is heated:

- 1 The submerged depth h of the block increases.
- 2 The upthrust (buoyant force) acting on the block decreases.
- 3 The liquid pressure acting on the bottom surface of the block remains unchanged.

Which of these statements is/are correct?

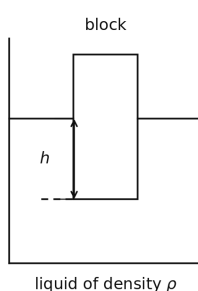


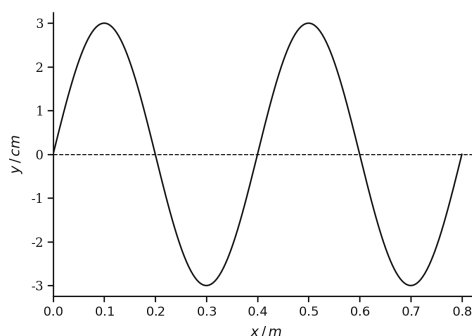
Diagram not to scale

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3
- 2 A radioisotope thermal generator uses a radioactive isotope P, which decays into a single stable isotope Q. The heat generated is proportional to the activity of the sample.
- The fuel is initially pure isotope P, which has a half-life of 20 years. After 80 years of operation, what is the ratio of the number of atoms of Q to the number of atoms of P?
- A** 1/16
- B** 1/15
- C** 7
- D** 15/16
- E** 15
- F** 16

- 3 A continuous transverse wave travels at a constant speed of 12 m s^{-1} along a stretched string in the positive x -direction. The graph shows the displacement y of the string as a function of position x at time $t = 0$.

What is the total distance travelled by a particle of the string during a time interval of 0.50 s , and what is the distance travelled by the wave during this same time interval?

	distance travelled by particle / m	distance travelled by wave / m
A	0.45	6.0
B	0.45	12.0
C	0.90	6.0
D	0.90	12.0
E	1.8	6.0
F	1.8	12.0
G	3.6	6.0
H	3.6	12.0



- 4 An electric heater supplies energy at a constant rate of 400 W to a metal block of mass 3.0 kg . The specific heat capacity of the metal is $1000 \text{ J kg}^{-1} \text{ K}^{-1}$. The block loses heat to its surroundings at a constant rate of 100 W .

What is the time taken for the temperature of the block to increase by 20 K ?

- A 120 s
- B 150 s
- C 200 s
- D 300 s
- E 600 s

- 5 Spherical catalyst particles of radius r fall through a gaseous reactant mixture. The drag force F acting on a particle moving with velocity v is given by:

$$F = kr^2v^2$$

where k is a constant.

The mass of a particle is proportional to r^3 . At the terminal velocity v_t , the drag force is equal to the gravitational force.

Which of the following gives the relationship between v_t and r ?

- A $v_t \propto r^2$
 - B $v_t \propto r$
 - C $v_t \propto \sqrt{r}$
 - D $v_t \propto \frac{1}{\sqrt{r}}$
 - E $v_t \propto r^{\frac{3}{2}}$
- 6 An astronaut of mass 100 kg is at rest in space. The astronaut throws a 4.0 kg tool, which then moves at a speed of 5.0 m/s. What is the astronaut's recoil speed?
- A 0.05 m/s
 - B 0.20 m/s
 - C 1.0 m/s
 - D 5.0 m/s
 - E 20 m/s
 - F 125 m/s

- 7 A U-shaped magnet rests on an electronic balance. A rigid, horizontal straight wire of length 15 cm is clamped independently in a fixed position between the poles of the magnet, perpendicular to a uniform magnetic field of flux density 0.40 T.

A constant current is switched on in the wire such that a total charge of 45 C passes through the wire in 2.5 minutes. This current causes a vertically upward magnetic force to act on the wire.

What is the resulting change in the reading displayed by the balance?

(Take the acceleration of free fall to be $g = 10 \text{ m s}^{-2}$.)

- A decrease of 0.18 g
- B increase of 0.18 g
- C decrease of 1.8 g
- D increase of 1.8 g
- E decrease of 18 g
- F increase of 18 g
- G decrease of 108 g
- H increase of 108 g

- 8 A circuit is set up as shown. The battery has negligible internal resistance and all three resistors are identical.

When the switch is open, the reading on the ammeter is 3.0 A and the power transferred from the battery is 18 W .

The switch is now closed.

What is the new reading on the ammeter and what is the new power transferred from the battery?

	ammeter reading / A	power transferred / W
A	1.5	18
B	1.5	24
C	2.0	12
D	2.0	24
E	2.0	36
F	3.0	24
G	3.0	27
H	3.0	36

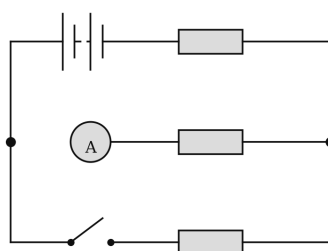
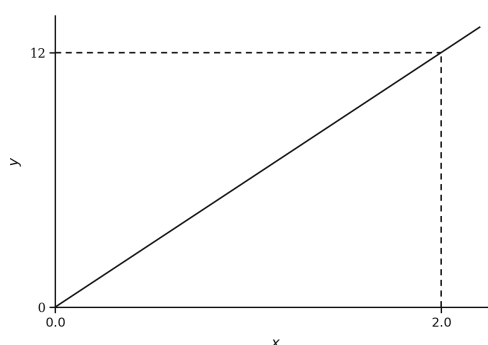


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- 9 The graph shown of quantity y against quantity x represents a physical relationship, where both quantities are measured in their standard SI units.

Which two of the following relationships could this graph represent?

- 1 momentum against velocity for an object of mass 6.0 kg
- 2 kinetic energy against velocity for an object of mass 12 kg
- 3 electrical energy transferred against charge transferred across a component with a constant potential difference of 6.0 V
- 4 tension against extension for a system of two identical springs connected in series, each having a spring constant of 6.0 N m^{-1}



- A** 1 and 2

B 1 and 3

C 1 and 4

D 2 and 3

E 2 and 4

F 3 and 4
- 10 A drone travels in a straight line between two points separated by a total distance of 800 m .
For the first distance d , it travels at a constant speed of 10 m s^{-1} . For the remainder of the journey, it travels at a constant speed of 25 m s^{-1} .
The average speed for the entire journey is 16 m s^{-1} .
What is the value of d ?

A 200 m

B 250 m

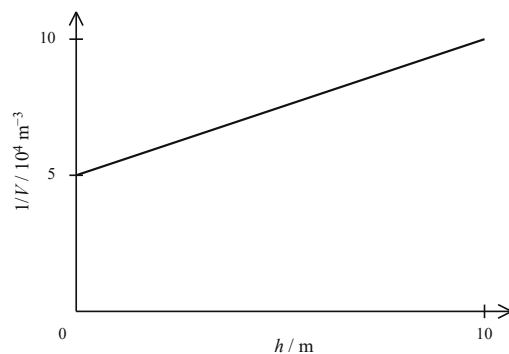
C 300 m

D 400 m

E 500 m

F 600 m

- 11 An air bubble rises through water of constant temperature. A graph of the reciprocal of the bubble's volume, $1/V$, against its depth, h , is a straight line as shown.



What is the atmospheric pressure at the surface of the water?

(Use: density of water, $\rho = 1000 \text{ kg m}^{-3}$; gravitational field strength, $g = 10 \text{ N kg}^{-1}$)

- A $1.0 \times 10^5 \text{ Pa}$
- B $2.0 \times 10^4 \text{ Pa}$
- C $5.0 \times 10^4 \text{ Pa}$
- D $2.0 \times 10^5 \text{ Pa}$
- E $1.0 \times 10^4 \text{ Pa}$
- F $5.0 \times 10^5 \text{ Pa}$

- 12 A uniform ladder of length L and weight W rests in static equilibrium with its upper end against a smooth vertical wall and its lower end on a rough horizontal floor.

The ladder is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{4}{3}$.

The coefficient of static friction between the ladder and the floor is $\mu = 0.50$.

A person of weight $2W$ begins to climb slowly up the ladder.

What is the maximum fraction of the length of the ladder that the person can climb from the base before the ladder begins to slip?

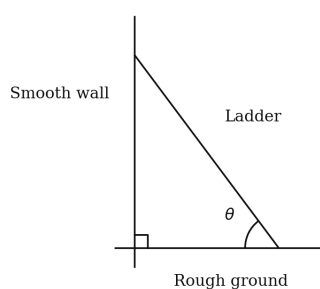


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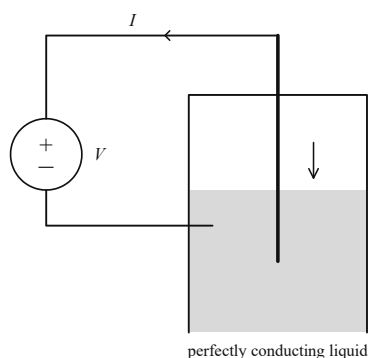
- A $\frac{5}{16}$
- B $\frac{1}{2}$
- C $\frac{2}{3}$
- D $\frac{3}{4}$
- E $\frac{7}{8}$

- 13 A student performs a calorimetry experiment using a shiny copper beaker containing hot water. To improve the accuracy of the enthalpy change determination, the student considers several modifications to reduce heat loss to the surroundings.

Which of the following modifications would be **counter-productive** (i.e., would increase the rate of heat loss)?

- A Coating the outer surface of the beaker with a thin layer of dull black paint
- B Polishing the outer surface of the beaker to a mirror finish
- C Placing a plastic lid on top of the beaker
- D Placing the beaker on a cork mat
- E Wrapping the sides of the beaker in mineral wool

- 14** The figure shows a circuit containing a power supply of constant voltage V and a uniform resistance wire. The wire is lowered at a constant speed into a bath of perfectly conducting liquid, shortening the length of the wire in the circuit. This process starts at time $t = 0$.



Which graph of current I against time t best represents this process?

- A** A graph showing current starting at a positive value and increasing with an increasing gradient (concave up).
 - B** A graph showing current starting at a positive value and increasing linearly (a straight line with positive gradient).
 - C** A graph showing current starting at a positive value and increasing with a decreasing gradient (concave down).
 - D** A graph showing current remaining constant at a positive value (a horizontal line).
 - E** A graph showing current starting at a positive value and decreasing linearly (a straight line with negative gradient).
- 15** A mineral sample contains a radioactive parent isotope X and its stable daughter product Y . The ratio of the number of nuclei of X to the number of nuclei of Y is $1 : 3$. Assuming that no Y was present initially and that X decays with a half-life of τ , what is the age of the sample?
- A** 1.5τ
 - B** 2τ
 - C** 2.5τ
 - D** 3τ
 - E** 4τ

- 16** Two identical, inverted test tubes, X and Y, contain trapped air. Tube X has a total mass m_X and tube Y has a total mass m_Y , where $m_X > m_Y$.

Both tubes are submerged in water and come to rest, neutrally buoyant, at depths h_X and h_Y respectively.

The temperature is constant throughout.

Which statement correctly compares the depths h_X and h_Y ?

- A** $h_X < h_Y$
- B** $h_X > h_Y$
- C** $h_X = h_Y$
- D** The relationship depends on the specific ratio m_X/m_Y .
- E** The relationship depends on the value of atmospheric pressure.
- F** There is not enough information to determine the relationship.

- 17** A resistor of constant resistance dissipates 36 W when the potential difference across it is 12 V.

The potential difference is changed to 8 V.

What is the power dissipated by the resistor at this new potential difference?

- A** 16 W
- B** 24 W
- C** 36 W
- D** 54 W
- E** 81 W

- 18** A number of different ions are all travelling at the same speed. They enter a region of uniform magnetic field, which is directed perpendicular to their velocity, and they subsequently travel in circular paths.

The ions are composed of protons (p), neutrons (n), and electrons (e) as listed below. Which ion follows the path with the largest radius?

- A** 20 p, 20 n, 18 e
- B** 20 p, 22 n, 18 e
- C** 20 p, 23 n, 18 e
- D** 20 p, 24 n, 18 e
- E** 20 p, 26 n, 18 e

- 19** An object of constant mass m is in motion. A graph of its kinetic energy against the square of its momentum (p^2) is plotted.

The graph is a straight line passing through the origin with gradient G .

What is the mass m of the object in terms of G ?

- | | |
|----------|----------------|
| A | $\frac{1}{2G}$ |
| B | $\frac{1}{G}$ |
| C | $\frac{G}{2}$ |
| D | $2G$ |
| E | $\frac{2}{G}$ |
| F | G |

- 20** In an AC circuit, the charge Q stored on a capacitor varies sinusoidally with time. The amplitude of the charge oscillation is Q_0 and the frequency is f .

What is the average magnitude of the current in the circuit over many complete cycles?

- A** 0
- B** fQ_0
- C** $2fQ_0$
- D** $4fQ_0$
- E** $2\pi fQ_0$

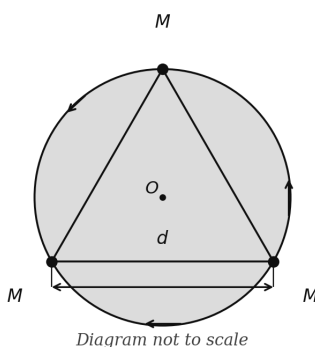
- 21** A wave travels from medium P to medium Q. In medium P, the wave speed is 400 m/s and the wavelength is 0.60 m. In medium Q, the wave speed is 1200 m/s.
- What is the wavelength of the wave in medium Q?
- A** 0.20 m
 - B** 0.60 m
 - C** 1.80 m
 - D** 5.40 m
- 22** A particle is projected vertically upwards on a planet. In the absence of an atmosphere, the particle would reach a maximum height H_0 .
- In the presence of an atmosphere, the particle experiences a constant resistive force of magnitude $\frac{1}{4}$ of its weight.
- Which expression gives the maximum height H reached by the particle in the presence of the atmosphere?
- A** $\frac{3}{4}H_0$
 - B** $\frac{4}{5}H_0$
 - C** H_0
 - D** $\frac{5}{4}H_0$
 - E** $\frac{4}{3}H_0$
- 23** A microwave transmitter and a detector are both positioned at a height of 5.0 cm above a large, flat, horizontal metal plate. The horizontal distance between the transmitter and detector is 24.0 cm.
- The detector registers a minimum in signal intensity.
- What is the maximum possible wavelength of the microwaves?
- A** 1.0 cm
 - B** 2.0 cm
 - C** 4.0 cm
 - D** 5.0 cm
 - E** 10.0 cm

- 24** A power supply has a constant EMF and a constant internal resistance, r .
 When the power supply is connected to a $3.0\ \Omega$ external resistor, the terminal potential difference is $9.0\ \text{V}$.
 When it is connected to a $12.0\ \Omega$ external resistor, the terminal potential difference is $12.0\ \text{V}$.
 What is the value of the internal resistance r ?
- A** $1.5\ \Omega$
 - B** $3.0\ \Omega$
 - C** $4.5\ \Omega$
 - D** $7.5\ \Omega$
 - E** $12.0\ \Omega$
 - F** $13.5\ \Omega$

- 25** Three identical stars, each of mass M , are located at the vertices of an equilateral triangle of side length d in deep space.

The three stars rotate in a common plane in a circular orbit about their center of mass, maintaining the equilateral triangle configuration with constant side length d , as shown in the diagram.

What is the orbital period T of each star in terms of the gravitational constant G , mass M , and distance d ?

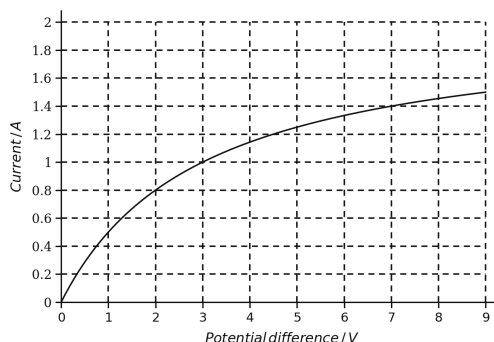


- A** $2\pi\sqrt{\frac{d^3}{GM}}$
- B** $2\pi\sqrt{\frac{d^3}{\sqrt{3}GM}}$
- C** $2\pi\sqrt{\frac{d^3}{3GM}}$
- D** $2\pi\sqrt{\frac{3d^3}{GM}}$
- E** $\pi\sqrt{\frac{d^3}{3GM}}$
- F** $2\pi\sqrt{\frac{d^3}{6GM}}$

- 26** A non-linear electrical component X has the current–potential difference ($I - V$) characteristic shown in the graph below.

Two identical components of type X are connected in parallel with each other. This parallel combination is connected in series with a $3.0\ \Omega$ resistor and an ideal d.c. power supply of electromotive force (EMF) $9.0\ \text{V}$ with negligible internal resistance.

What is the total power dissipated in the entire circuit?



- A** 3.0 W
 - B** 6.0 W
 - C** 9.0 W
 - D** 12 W
 - E** 15 W
 - F** 18 W
 - G** 24 W
 - H** 27 W
- 27** A stationary sound source emits a continuous note of frequency $3300\ \text{Hz}$. A stationary observer is positioned some distance away from the source.
- A wind blows directly from the source to the observer at a constant speed of $30\ \text{m/s}$. The speed of sound in still air is $330\ \text{m/s}$.
- What is the frequency of the sound detected by the observer?
- A** 3000 Hz
 - B** 3025 Hz
 - C** 3300 Hz
 - D** 3600 Hz
 - E** 3630 Hz